

Lab No	TOPICS	Course Outcome Addressed
Lab 1	Demo 1: Python Language Basics Exercises 1. Write a Python function to input two numbers and perform the Calculator operations of (+, -, *, /). 2. Write a Python function that takes an integer and returns True if it's a prime number and False otherwise. 3. Create a Python function that creates a sequence between 1 and 100 and prints all the odd numbers. Compute and display the sum of all the even numbers. 4. Write a Python function to add two elements and display the result. The elements can be of type integer, float or string. 5. Write a Python function that takes a string input from the user and counts the number of vowels and consonants in the string.	CO1
Lab 2	Demo 2: Python built-in Data structures, Functions, modules, packages Exercises 1. Write a Python code block that inputs numbers into a list. Print the largest, smallest, the sum, and the average of the numbers. Count occurrences of a specific number in the list. 2. Write a Python code block to create a tuple with five elements. Try to change one of the elements and handle the error that occurs. Print a message that explains why the error occurred. 3. Write a Python code block to create a dictionary of cricket World Cup winners. Let the key be the year; the value is the country that won the World Cup that year. Print the name of the best-performing country. Display the unique list of countries that have won the World Cup. 4. Write a Python code block that inputs a sentence from the user. Count the frequency of each word in the sentence and store the result in a dictionary. Prints the dictionary with words as keys and their frequencies as values. 5. Write a Python code block to input numbers into two sets. Perform union, intersection, and difference operations on the sets and print the results.	CO1
Lab 3	Demo 3: NumPy basics and vectorized computation Exercises 1. Generate a 3x4 NumPy array with random integers between 1 and 50. a. Calculate and print the Mean, Median, and Standard Deviation of the array b. Print the Sum of all elements and the sum of each row. c. Reshape the 3x4 array into a 2x6 array and print it.	CO2

	2. Create two (3 * 3) matrices using NumPy and print it. Perform and print the results of the following linear algebra operations a. Matrix addition b. Matrix subtraction c. Matrix multiplication (element-wise and dot product) d. Transpose of a matrix e. Determinant and inverse (if applicable)	
Lab 4	Demo 4: Pandas, Data loading, Storage and File formats Exercises 1. Create a Series from a list of integers representing daily temperatures (in Celsius) over a week. Assign index labels as day of the week. a. Find and print the average (mean) temperature for the week. b. Identify and print the maximum and minimum temperatures and their respective days. c. Display the temperatures greater than a specific value. d. Convert all temperatures to Fahrenheit. e. Print the days had temperatures above the average. 2. Create a data frame with details of 10 students and columns as Roll Number, Name, Gender, Marks1, Marks2, Marks3. a. Create a new column with total marks b. Find the lowest marks in Marks1 c. Find the Highest marks in Marks2 d. Find the average marks in Marks3 e. Find student name with highest average f. Find how many students failed in Marks2 (<40)	CO2
Lab 5	Demo 5: Data Cleaning and Preparation 1. Create a CSV file called "Movies.csv" with details of 10 movies- Movie Name, Language, Genre, Rating, Review. a. Read CSV file into a dataframe and find the movie with the highest rating. b. Write the details of all "Hindi movies into a file "HindiMovies.csv". 2. For the CEREALS dataset, perform data preprocessing and answer the following questions. a. Create a table with the 5 number summary of all the numeric attributes. b. For each of the numeric attributes (proteins upto vitamins) , identify and replace all missing data(indicated with -1) with the arithmetic mean of the attribute. c. Create a table with the 5 number summary of all the numeric attributes after treating missing values. Do you think the strategy used in dealing with missing values was effective? d. For each numeric attribute (proteins upto vitamins), identify and replace all noisy data with the median of attribute. e. Create a table with the 5 number summary of all the numeric attributes after treating noisy values. Do you think the strategy used in dealing with noisy values was effective?	CO3

Lab 6 – Demo 6: Data Visualization, Context & Storytelling

Dataset: MTCARS & CEREALS

Lab Questions (MTCARS Dataset)

1. Load the **MTCARS** dataset and display its basic structure (rows, columns, data types).
2. Compute summary statistics (mean, median, min, max) for:
 - Miles per gallon (mpg)
 - Horsepower (hp)
 - Weight (wt)
3. Identify the relationship between engine cylinders (cyl) and fuel efficiency (mpg) using an appropriate visualization.
4. Create a scatter plot between horsepower (hp) and mpg.
 - Add labels and a meaningful title.
5. Compare manual vs automatic transmission cars in terms of mpg using a box plot.
6. Identify which car model gives the best balance of power and fuel efficiency. Justify using a visualization.
7. Write a short data story (3–4 lines) summarizing insights derived from the plots.

Additional questions (CEREALS Dataset)

1. Load the **CEREALS** dataset and inspect missing values.
2. Calculate average calories, sugar, and fiber content.
3. Create a bar chart showing top 5 cereals with highest sugar content.
4. Visualize the relationship between calories and rating.
5. Compare cold vs hot cereals based on calorie distribution.
6. Identify healthier cereals using at least two nutritional parameters.
7. Summarize findings using one key visualization and explanation.

Lab 7 – Demo 7: Plotting using Matplotlib & Seaborn

Dataset: IPL

Lab Questions

1. Load the **IPL dataset** and display first 10 records.
2. Find the total number of:
 - Matches
 - Teams
 - Seasons
3. Plot the number of matches played per season using Matplotlib.
4. Use Seaborn to visualize:
 - Top 5 teams with most wins
5. Create a bar plot showing top 10 run scorers.
6. Plot a pie chart representing match wins by venue.
7. Compare home vs away wins using an appropriate plot.
8. Create a visualization that tells a clear IPL performance story (your choice).
9. Add annotations, legends, and titles to improve readability.

Lab 8 – Demo 8: Data Aggregation & Group Operations

Dataset: NORTHWIND

Lab Questions

1. Load the NORTHWIND dataset and explore all available tables.
2. Calculate:
 - Total sales per product
 - Total sales per category
3. Identify the top 5 customers by revenue.
4. Group data by country and compute total orders.
5. Find the monthly sales trend using aggregation.
6. Perform a groupby operation to calculate average order value per employee.
7. Create a bar chart showing category-wise sales.
8. Visualize country-wise revenue contribution.

9. Interpret results and write 2 business insights based on aggregation.

Lab 9 – Demo 9: String Manipulation & Data Wrangling

Dataset: SENTIMENT

Lab Questions

1. Load the SENTIMENT dataset and inspect text columns.
2. Perform basic string operations:
 - Convert text to lowercase
 - Remove punctuation
3. Count the number of positive, negative, and neutral sentiments.
4. Find the most frequently occurring words in positive reviews.
5. Remove stopwords and display cleaned text.
6. Create a word length distribution plot.
7. Generate a bar chart showing sentiment count.
8. Identify top keywords influencing negative sentiment.
9. Write a short summary explaining how string processing helped sentiment analysis.